Feasibility Validation Report - Codefi Farm Financial Planning

Executive Summary

Feasibility Assessment: FEASIBLE with HIGH CONFIDENCE

Key Finding: MVP is technically achievable in 6 weeks with 1–2 engineers and $30–35K budget. Success is conditional on closing 2 co-op LOIs by Week 2 to validate go-to-market channel. Unit economics hold at 5.1:1 LTV:CAC with hybrid LLM approach and co-op partnerships.

MVP Timeline Estimate: 6 weeks to market; Month 2–4 validation gates determine scaling strategy

Confidence Level: HIGH based on: (1) constrained technical scope, (2) validated customer willingness-to-pay, (3) realistic operational model with agronomist-assisted onboarding, (4) no regulatory blockers, (5) $425K seed capital aligns with build + pilot + team costs

Technical Feasibility

Core Technology: - Stack: Python/Node backend, React frontend, PostgreSQL, AWS infrastructure, OpenAI API (future, not MVP) - Complexity: LOW-MEDIUM. Rules-based scenario engine (deterministic math, no AI in MVP) - Proven: YES. Excel-like calculations, CSV data import, PDF export are industry-standard patterns

Technical Risks & Mitigations: - Data quality gate: Progressive disclosure form (3 required fields, optional below) prevents garbage-in-garbage-out. Validation rules enforce numeric ranges ($5–50/seed unit, $200–800/fertilizer/acre). Mitigation: Aggressive inline validation, not post-hoc cleanup. - LLM cost scaling: Hybrid approach uses LLM for natural language parsing only; rules engine handles math. Reduces token costs 80%, keeps usage under $50/month at break-even scale. - Co-op data integration complexity: Assume CSV import for MVP (1 week), not automated API. Validates data format and partnership appetite before Week 8 automation.

Scalability Assessment: Architecture scales to 1,000+ customers with managed PostgreSQL and standard AWS auto-scaling. No architectural rewrites needed through Year 2.

Operational Feasibility

Key Operations: - Farmer onboarding: Agronomist-assisted via co-op partners (reduces direct CAC, increases activation). Co-ops provide human touchpoint; you provide tech support. Medium complexity, proven model in ag sector. - Scenario modeling: Rules-based calculation (no bottleneck). Farmer inputs → P&L in <2 seconds. Low complexity. - Data aggregation for benchmarking: SQL aggregations (percentiles, outliers, regional stratification). Standard data warehouse work. Medium complexity, well-understood.

Resource Requirements: - Team: 1 full-time engineer (6 weeks MVP) + 1.5 part-time engineers (infrastructure, QA) + 1 co-founder/operator (co-op partnerships, pilots). Add 1 CS person by Month 3 for pilot support. - Tools: AWS ($500/mo), Stripe ($0 until revenue), Auth0 ($0 free tier), Mixpanel ($200/mo). Security audit ($5–10K, one-time).

Bottlenecks Identified: - Co-op data access: Conditional on signed LOIs by Week 2. If fails, pivot to assisted onboarding (adds $150K/year CS cost, increases direct CAC). Mitigation: Contact 3 co-ops Week 1; have 2 LOIs by Week 2. - Cost entry friction: Farmers may spend 40+ minutes entering data (not your modeled 15). Mitigation: Agronomist help during onboarding; pre-population from co-op CSV reduces friction by 60%. - Benchmarking dataset: Need 50+ farms before statistical significance. Delayed to Week 8 post-MVP. Pilot farmers see 'coming soon' until then. Risk: Churn before benchmarking launches. Mitigation: Transparent roadmap + early-adopter messaging.

Build vs Buy Analysis

Build (MVP): - Scenario engine with rules-based P&L math (proprietary logic, 80 eng hours) - Cost entry form + validation (60 hours) - Decision logging + PDF export (60 hours) - CSV data import (manual upload, 20 hours)

Buy/Partner: - LLM API: OpenAI GPT-4 (parse input for future; skip MVP). Cost: $0.01–0.03 per scenario at scale. - Cloud infrastructure: AWS managed services (RDS, AppServer, S3). Cost: $500/mo MVP, scales linearly. - Auth/Identity: Auth0 free tier (100 users). Upgrade: $1,200/year at 1K users. - Payment: Stripe (0% until revenue, then 2.9% + 30¢ per transaction). - PDF library: ReportLab open-source, no cost. - Analytics: Mixpanel ($200/mo, alternative: Segment $500/mo).

Partner (Not Buy): - Co-op data integration: Negotiate data sharing agreements (DPA, CSV export schedule). Build custom adapters per co-op (1 week each). No third-party platform purchase. - Lender integration: Manual scenario export (Week 6), not API integration (defer to Month 3+).

Integration Complexity: CSV import adds 1 week to MVP. Automated API integrations deferred to Week 8+ pending co-op technical discussions. DPAs (legal, not technical) require 1 week per co-op parallel to engineering.

Compliance & Regulatory

Regulatory Requirements: - No lending/investment licenses: Tool is decision-support, not financial advice. No regulatory barriers. - Data privacy (GDPR/CCPA): Farm financial data is sensitive. Mitigation: US-only data residency (AWS us-east-1), encrypted at rest/in transit, farmer consent for benchmarking, transparent privacy policy. - Data Processing Agreements: Co-ops control farmer data; you are processor. 1-week legal template per co-op.

Data Privacy Approach: - Farmer opt-in for benchmarking: 'Your anonymized costs help peers. Consent here.' Transparent, not opt-out. - Anonymization rigor: Ensure 5+ farms per benchmark cell (prevents de-anonymization). Strip PII (names, coordinates; use county/region only). Third-party anonymization audit: $5K, Week 6. - Access logging: CloudWatch + DataDog track who accessed which farm data, when. Cost: $2K tooling, 2 weeks implementation.

Compliance Roadmap: - Week 5–6: Third-party security audit ($5–10K). Covers source code, infrastructure, data handling. Delivers proof for pilots. - Week 6: DPA templates signed with co-ops (legal, 1 week per partner). - Year 2: SOC 2 Type II optional (if scaling past 100 customers); cost ~$15–25K, 3–6 month process.

No blockers. Compliance is standard SaaS hygiene, not novel risk.

Expert Debate Insights

Round 2 Key Debates:

Direct vs. Co-op Channel: Panel split on whether to pursue both in MVP. Resolution: Focus 80% on co-op partnerships (agronomist-assisted onboarding, pre-populated data). Direct channel CAC is $1,500+ without data pre-pop (unit economics break). Defer direct to Month 6+ as pivot if co-op adoption < 2%.

LLM Cost Scaling: Concern: At 1,000 customers × $0.01/scenario × 10 scenarios/month = $100/month in tokens. Unit economics assume $127/month margin. Resolution: Hybrid approach (LLM parses input, rules engine runs math) cuts token usage 80%, keeping costs under $20/month at scale.

Co-op Integration Complexity: Unknown unknowns on co-op system architecture. Resolution: Contact IT directors Week 1 to scope integration (CSV import for MVP, API for Year 2). Don't build automated integrations until you know partner data format.

Consensus: Technical scope is achievable. Success hinges on co-op partnerships closing by Week 2 and data entry friction being manageable with agronomist assistance. Cost entry validation must be aggressive to hit Month 2 gate (40%+ farmers complete entry in <30 minutes).

Implementation Roadmap

Week 1–2: Foundation + Co-op Validation - Engineering: AWS setup, cost form skeleton, scenario engine rules definition (60 hrs) - Operations: Contact 3 co-ops; close 2 LOIs with scoped data integrations by Week 2 (YOUR critical path) - Compliance: Draft DPA templates, finalize farmer consent language

Week 2–4: Core Build - Engineering: Cost entry + validation, scenario engine, decision logging (180 hrs) - Operationss: Agronomist coordination, pilot recruitment materials, data export testing

Week 4–6: Integration + Launch - Engineering: CSV upload, analytics instrumentation, security audit remediation (80 hrs) - Operations: 10–15 soft-launch with pilot farmers (Week 5); hard launch 80–125 farmers (Week 6) - Compliance: Security audit delivery + sign-off

Post-MVP (Month 2–4): - Month 2 Gate: 40%+ farmer activation (cost entry + scenario run + decision log) - Month 3: Launch benchmarking with pilot cohort data - Month 4 Gate: 60%+ retention, clear channel winner identified - Month 6: Scale winning channel, 40–60 paying customers

Recommendations

Immediate Actions (Week 1): 1. Contact 3 co-op IT + legal directors: Ask scope of data integration (CSV vs. API), data format, export frequency. Have 2 LOIs signed by Week 2. 2. Hire security audit firm: Scope SOW for source code + infrastructure review. Cost $5–10K. Schedule delivery Week 5–6. 3. Finalize MVP feature prioritization: Scenario engine + cost entry + decision logging ship Week 6. Defer benchmarking (Week 8), lender API (Month 3), commodity feeds (Month 4+).

Before Building: - Lock co-op partnerships with signed data agreements (gates entire pilot distribution) - Confirm agronomist availability for co-op onboarding support (operational, not technical) - Define validation rules for cost entry (numeric ranges, required vs. optional fields)

Team Priorities (Months 1–2): - Hire 1 full-time engineer (6-week MVP build). 1.5 part-time engineers for infrastructure/QA. - Add 1 part-time ops person (data validation + benchmarking dataset prep during pilot). - Plan customer success hire by Month 3 (pilot support, agronomist coordination).

Budget Allocation: - Engineering: $24K (240 hours @ $100/hr) - Cloud/Tools: $1.4K (infrastructure + audit firm + software subscriptions) - Total MVP: $35K. Remaining: $390K for team (3–4 people), operations, co-op pilots, marketing.

Critical Success Factors & Risk Mitigation

Make-or-Break Milestones:

1. **Week 2: Co-op LOIs closed** – If 2+ co-ops sign partnerships with data access, proceed to Week 3. If fails, pivot to direct-only channel + assisted onboarding (adds $150K/year CS cost).

2. **Month 2: Data entry friction < 40 min** – If 40%+ farmers complete cost entry in <40 minutes, CAC model is viable. If <30%, implement agronomist-assisted entry and pre-population from co-op CSV.

3. **Month 4: 60%+ retention + scenario quality** – If farmers retain and report scenarios match intuition, scale validated channel. If churn or poor quality, redesign cost entry or scenario output clarity.

Key Risks & Mitigations:

Risk	Probability	Mitigation
Co-op partnerships fail	Medium	Contact 3 co-ops Week 1; have 2 LOIs by Week 2. Pivot to direct + paid support if both fail.
Cost entry takes 60+ min	Medium-High	Agronomist assist + CSV pre-population reduces to 20–30 min. Progressive disclosure UX minimizes friction.
LLM costs spiral at scale	Low	Hybrid rules-engine approach caps tokens at $20/month/100 customers.
Security audit finds blockers	Low	Scope known risks upfront; fix before Week 6 launch.
Farmers churn before benchmarking	Medium	Launch Week 8 with pilot data. Communicate roadmap transparently.

Accept/Defer Risks (Not MVP): - Mobile experience suboptimal (desktop-first acceptable for pilots) - Lender integrations manual (PDF export sufficient Year 1) - No real-time commodity feeds (farmer manual input acceptable)